

POSTnote 777

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Digital sovereignty



Summary

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- 2 How sovereign is the UK?
- 3 How can UK digital sovereignty be increased?

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Summary

Digital sovereignty is the agency and capacity a country or organisation has to shape, use and maintain the digital systems it depends on. These systems consist of three layers: infrastructure (such as data centres), code (including artificial intelligence), and data (such as health records).

If countries, companies and citizens can employ digital sovereignty they may be able to direct the use of digital systems to achieve a range of aims, including security and resilience, economic growth, climate commitments, and democratic accountability.

However, there can be risks. If countries adopt overly 'protectionist' approaches it could restrict trade, reduce access to leading technologies, and weaken international collaboration.

Stakeholders say that digital sovereignty does not mean isolationism or complete self-sufficiency. Instead, they say countries should exercise digital sovereignty by identifying and reducing their dependency on others for their most critical digital technologies, while also working with a wide range of international partners. They say this can reduce risks by diversifying who the country depends on, meaning the dependencies are less concentrated.

How sovereign is the UK?

The UK Government has taken steps to strengthen sovereignty through investments, including in AI, semiconductors, and sovereign timing capabilities (which are needed, for example, for accurate geolocation and coordinated financial transactions). In May 2026, the Competition and Markets Authority (CMA) launched a strategic market status investigation into a large software vendor to see if its business software reduces competition in the cloud services industry.

The UK Government is also forming plans for national payment systems (almost all card payments in the UK are currently administered by US companies Visa and Mastercard).

The UK also has strengths that could be used to improve sovereignty, such as advanced domestic capability in semiconductor design, research excellence, and manufacturing compound semiconductors. UK organisations such as the NHS and BBC also hold advanced datasets that could be used for training AI models.

However, the UK still has dependencies, for example in cloud providers, AI, advanced semiconductor manufacturing, and payment networks, without having a strategy for digital sovereignty that could address them.

For more detail on the models of sovereignty pursued globally, and comparison of EU and UK policy, please see the 2026 [House of Commons Library briefing](#).

How could UK digital sovereignty be increased?

Stakeholders say that to increase its digital sovereignty the UK needs clear evidence about which digital systems are critical, how supply chains work to support them, what they depend on, and how they could fail.

Some also say the UK could make better use of open-source software, freely available code that anyone can see, modify, use and distribute. They say this may help reduce 'vendor lock in', a dependence on a specific supplier that makes it difficult or costly for customers to switch to an alternative.

Parliamentary committees and the National Audit Office said that public procurement could do more to support UK digital sovereignty. They said the £14 billion of public funds spent on digital programmes and technology each year could be used to buy domestic products.

In 2023, US firm Palantir was awarded a £330 million contract to provide a data platform for the NHS. In 2026, the Science, Innovation and Technology Committee said that the UK Government should use the 2027 break clause in the contract and develop a UK-based data platform instead.

Some MPs have called for a sovereign procurement strategy.

Skills development may be important for digital sovereignty to ensure that the UK continues to have the skills to buy, check, operate and fix critical technologies.

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Background

The UK is increasingly [reliant on digital technology](#). In 2025, the UK Government estimated that the UK digital economy was worth £286 billion in gross value added (GVA), or 13% of national GVA.¹ Due to the significant impact on the UK economy and society, in 2026 the UK Government said “we must shape this technology, not just be shaped by it”.²

Digital sovereignty is the agency a country has to shape this digital world, which includes infrastructure (such as data centres), code (including artificial intelligence) and data (such as health records).

Interest in digital sovereignty in the UK and EU is driven by growing awareness of dependencies on certain technologies, a lack of competitiveness, vulnerability to [cyber threats](#), geopolitical insecurity, and emerging technology, such as AI.²⁻⁶

For example, the UK is dependent on many US technologies, including operating systems, payment cards, and AI.⁷⁻¹¹ In 2026, MPs raised concerns about dependency on US data analytics company Palantir for NHS systems.¹¹⁻¹³ Sanctions on International Criminal Court personnel in 2025 have also shown that access to US digital systems can be removed due to political disputes.¹⁴⁻¹⁶

However, some stakeholders, such as the CEO of industry body OpenUK, are concerned that discussions around sovereignty could be politically “weaponised” to exclude countries like the US.¹⁷

The digital world is highly interdependent,^{4,18} with reliance on countries beyond the US. For example, the [semiconductor supply chains](#) that underpin digital technology rely on Taiwan for the latest semiconductor chips, and the Netherlands for the machines to make them.¹⁹

In 2026, some MPs called for a digital sovereignty strategy,^{5,6} but the UK Government said that “sovereignty is complex and multifaceted”.²⁰

This POSTnote examines what digital sovereignty is, its risks and benefits, how it is being addressed, and examples to increase digital sovereignty in the UK. It does not discuss military implications. For more detail on the models of sovereignty pursued globally, and comparison of EU and UK policy, please see the 2026 [House of Commons Library briefing](#). For more detail on the resilience of cyber infrastructure, see the [House of Commons briefing on data centres](#), the [POST briefing on data centres](#), and also the [POST briefing on the cyber resilience of UK digital infrastructure](#).

1 What is digital sovereignty?

1.1 Definitions

There is no universally agreed definition of digital sovereignty. While there are common themes to the definitions in table 1, there can be different ways to apply them:

- Sovereignty is about control, but views vary on how much control is useful or desirable. Some say it is the agency and capacity to decide how to develop digital technology, and having the skills, knowledge and processes to do it well.^{21,22} It is widely agreed that isolationism or full self-sufficiency would be infeasible or undesirable.^{a 2,23–28}
- Digital systems have multiple layers to control, for instance:^{22,29}
 - physical: such as resources, infrastructure and hardware
 - code: such as software, standards, rules and system design
 - information: such as data and content
- Digital sovereignty may be a goal for countries, or for other entities such as individuals or companies. According to Civo's 2026 survey of 1,000 UK IT leaders, digital sovereignty is a priority for 73% of organisations.^{30–32} Companies such as SAP, IBM and BT divide digital sovereignty into:^{31,33,34}
 - technological sovereignty: retaining choice and flexibility in infrastructure^b
 - operational sovereignty: maintaining control over strategic decisions, meaning a company can run, support and build upon its technology^c
 - data sovereignty: controlling who has access to data

^a For example, the director of GCHQ said in 2026: "sovereignty doesn't have to mean made in the UK, so long as we carefully manage our supply chains, dependencies, and data".²³

^b Some experts consider 'technological sovereignty' to be just the physical layer,^{35–37} but it is also used interchangeably with 'digital sovereignty'.^{5,37}

^c Colin Bannon CTO of BT business says operational sovereignty asks the question "Can I keep my business running?".³⁸

Table 1: Example definitions of digital sovereignty	
Source	Definition
Public Digital (consultancy) ³⁹	"The agency and capacity of any organisation to make intelligent, informed choices to shape its digital future by design"
Open Rights Group (non-profit digital campaign organisation) ²⁵	"The ability of a country to have control over its digital infrastructure, data, and technology."
World Economic Forum (non-profit organisation) ³⁶	"The ability to have control over your own digital destiny – the data, hardware and software that you rely on and create."
Gartner (research and advisory firm) ³⁵	"An organization's ambition to maintain control over its data, operations, and technology in the face of external influences, ensuring continued innovation and resilience."
European Commission ⁴⁰	"Capacity to exercise strategic independence in the digital domain while remaining open and connected to global networks."
GCHQ (Government Communications Headquarters) director Anne Keast-Butler ²³	"The agency, ability and agility of nations to shape their own digital future."

How do other countries see digital sovereignty?

Table 2 shows the range of ways digital sovereignty is being approached around the world.^{36,41}

Table 2: Approaches to digital sovereignty around the world	
US	Does not use the term digital sovereignty, but advocates for an unregulated environment where 'big tech' plays a dominant role. ^{25,36}
China	Has a government-led approach to digital sovereignty. ³⁶ A large domestic market allows domestic companies to grow internally before going to international markets. ⁴² Some say China aims for a 'full stack' in AI, meaning it attempts to have capability across the whole technology supply chain. ⁴³
EU	Attempts to balance benefits of the market and rights of citizens, for example, GDPR. ⁴² Even within the EU, countries differ in how they seek to achieve digital sovereignty, and in their relationships with US technology. ⁴⁴ In 2026, the EU announced a technology sovereignty package to strengthen its capacity in semiconductors, AI, cloud services and open source software. It includes two pieces of legislation covering chips, cloud and AI, as well as a strategy for open source. It has a roadmap for integrating digital systems such as AI with the energy sector. ^{45–47}
African countries	Stakeholders suggest that some African countries in particular are concerned with data sovereignty. ^{41,48–50} For example, some countries are building data centres, such as the Republic of Congo (financed by public funds and the African Development Bank) and Senegal (financed by a Chinese loan). ^{41,51–54} In 2024, the Centre for International Governance Innovation said that "external countries [who provide funding] often dictate Africa's digital priorities". ⁴¹

What is the UK interpretation of digital sovereignty?

There is no official UK Government position on what digital sovereignty is.^{22,55,50,56}

The Minister for Digital Government and Data, Ian Murray, said in 2025 that the UK Government addresses sovereignty by investing in technology that is "physically here" in the UK.⁵⁷

In April 2026, the Secretary of State for Science, Innovation and Technology said that we will "welcome inward investment" and work "more closely with our international partners", recognising that sovereignty is not about "go[ing] it alone".²

Stakeholders say that clarity on a government definition and position would be helpful.^{11,56,58,59} Some have called for a digital sovereignty strategy.^{6,16}

1.2

What are the benefits and risks of digital sovereignty?

Digital sovereignty may have benefits beyond increasing security and resilience, and it may also have risks, for example [cutting off UK access to the best technologies](#). Views vary on what digital sovereignty should do, with objectives often overlapping.²²

Security and resilience

Digital sovereignty attempts to make the UK less vulnerable to digital threats, including cyberattack, espionage, supply chain issues, and withdrawal of services provided by major digital service providers.^{d 5,14,15,25,30}

However, one researcher warned that reduced international collaboration could mean global digital outages, such as the 2024 CloudStrike update, may not be resolved as quickly.^{e 62–65}

Economic growth

Some stakeholders say there are wider economic benefits to sovereignty, such as increased trust in sovereign AI (leading to greater adoption), a growing sovereign cloud market, and increased resilience, for instance, so that businesses can still operate during global IT outages.³¹

Others, such as the Open Rights Group, have concerns that dominant foreign technology companies may increase 'vendor lock in' (a dependence on a specific supplier that makes it difficult or costly for customers to switch to an alternative), which may increase costs for governments or companies.²⁵

At the same time, some approaches to sovereignty could pose risks to economic growth. Sabina Ciofu from techUK said there is a risk to trade if sovereignty becomes "a synonym for protectionism".^{66,67} For example, if the UK excludes foreign companies, then foreign countries may exclude the UK from their markets.²⁸

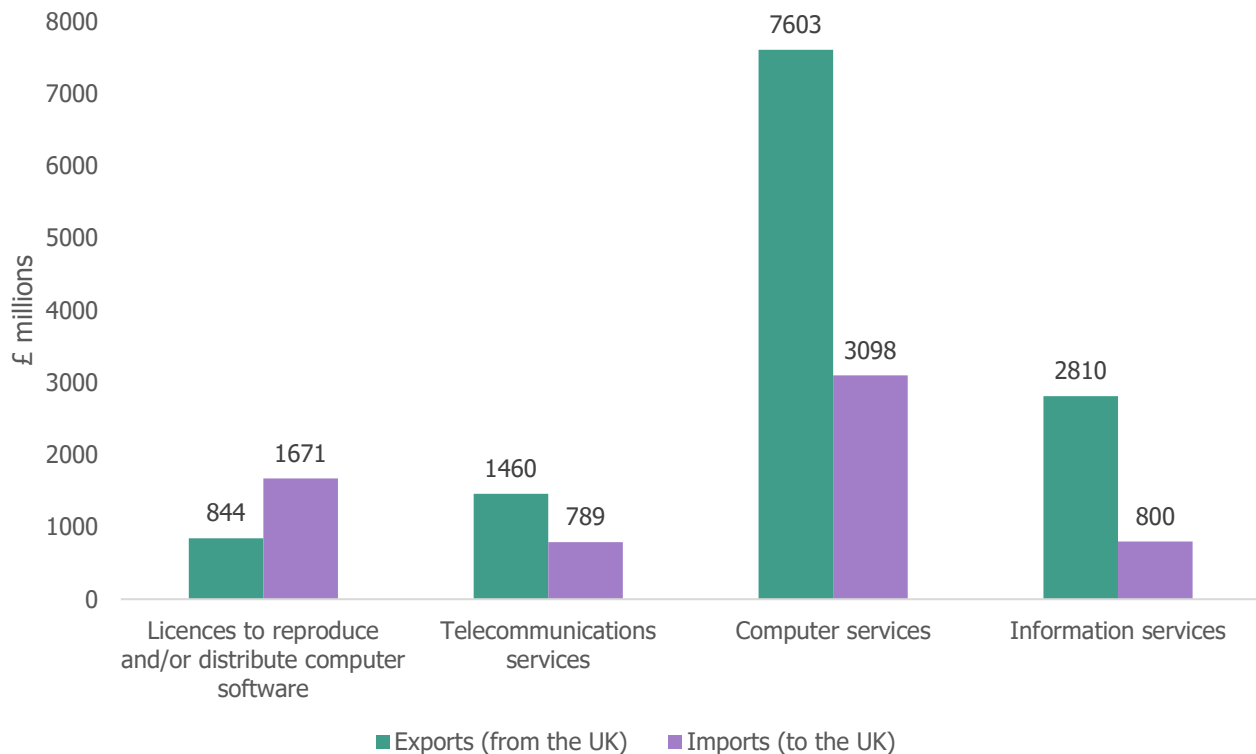
While many raise concerns of UK dependency on the US, ONS data shows that the UK is a net exporter of goods and services. For example, in 2025 the

^d In 2025, Karim Khan, Chief Prosecutor of the International Criminal Court (ICC) was Sanctioned by the US. As a result, his Microsoft email account was suspended, although a Microsoft director has said "it was the ICC's decision to terminate and to respond to that [...] sanction, not Microsoft's. So we were not [...] put in a position where we were terminating the agreement. We were working with the ICC throughout."^{60,61} Sanctions were extended to other ICC personnel later in 2025. Sanctioned French ICC judge, Nicolas Guillou, has described the "extent of disruption" he experienced, including not being able to use a US credit card or hotel reservation websites.^{14,15}

^e CloudStrike is a company that produces antivirus software. An update to their software caused widespread global IT failures in 2024, impacting flights, emergency services, TV broadcasting and financial services, among others.

UK exported £7.6 billion of computer services to the US, while importing £3.1 billion (figure 1).⁶⁸

Figure 1: Examples of UK imports and exports of services from the US in 2025



Source: Data from the Office for National Statistics (UK trade in services: service type by partner country, non-seasonally adjusted).⁶⁸

Preserving democracy

'Big tech' companies have increasing political influence. Clive Lewis MP said that algorithms have the "power to shape public debate", but with limited public oversight.^{25,69–73} The University of Cambridge AI initiative said that preserving democracy means ensuring that digital systems promote UK values, serve its citizens, and are subject to democratic oversight.²⁷

Climate commitments

Researchers say that compute infrastructure^f for digital sovereignty risks undermining climate commitments due to its [power demand](#).²⁷

However, digital sovereignty could encourage domestic industry to focus more on low-energy computing and hardware [circularity](#) to ensure there is sufficient energy and material available for sovereign capabilities.⁷⁸ Similarly, if semiconductor manufacturing moved into the UK, the UK could better

^f Compute can be a generic term for the resources required by computer programmes, for example processing power and storage.⁷⁴ Although some use it just to refer to the processing power.^{75–77}

control its environmental impact, for example, by requiring it to use more clean electricity.⁷⁹

2

How sovereign is the UK?

The UK is dependent on a complex mix of global technologies. It has opportunities to increase its capability at home and increase sovereignty. This section looks at some areas where the UK could increase its digital sovereignty or where it is taking some action.

2.1

Sovereign AI

UK interest in sovereign AI includes concerns about dependence on foreign technology providers, access to compute and data, and desire to have access to the latest technology.

UK Compute

In 2025, the UK Prime Minister said the UK should be an “AI maker, not just an AI taker”. He announced a £1 billion investment to ‘scale up’ the computing power available for AI training.^{9 80} This was expanded to £2 billion in the 2025 UK [Compute Roadmap](#), including a £750 million investment in an Edinburgh supercomputer.⁷⁵

In 2026, the government announced funding for chip development with the £1.1 billion [AI Hardware Plan](#).^{81–83}

In the Compute Roadmap, the government said the UK has a “fragmented system in need of direction”.⁷⁵ In 2026, the Tony Blair Institute said that sufficient compute is needed to develop AI, and to run AI for critical services during global disruption in compute access.²⁶

The Compute Roadmap said the UK needs 6 GW of AI-capable datacentre capacity by 2030.^{h 75} The government estimated UK datacentre capacity to be 1.6 GW in autumn 2024,⁸⁴ and in 2026, the Tony Blair Institute said at a committee hearing that the UK is not on track to hit the target.⁸⁵

⁹ See [POSTbrief 57](#) for an information on the ways AI can be trained, the training data needed and the energy needed to train AI.

^h In comparison, the UK used around [36 GW of electricity on average in 2024](#).

Box 1: The Sovereign AI Fund

The Sovereign AI fund, launched in 2026, will allocate £500 million in venture capital and research and development (R&D) funding to UK AI startups.⁸⁶ It also offers additional support that the chair of Sovereign AI, James Wise, said is worth “a lot more” than the funding. This includes:⁸⁷

- funded access to compute
- access to public procurement opportunities
- funding for visas for R&D talent
- grants to build AI assets such as datasets

The fund targets specific niches to build on UK strengths, and supplement the £6 billion of UK AI private venture capital in 2025. The chair said it is expected to make returns on investment for taxpayers.⁸⁸

The AI Minister Kanishka Narayan said that the fund intends to keep companies “anchored in the UK”, but there are no “strict conditionalities” on investments.⁸⁹

Can the UK build its own frontier AI?

Frontier AI models are the most advanced general purpose models, with some costing billions of pounds to develop.⁹⁰

In 2026, the Tony Blair Institute said that most countries will not be able to build frontier AI (outside the US or China) due to the cost, compute, data, and talent required. They said that countries unable to build their own could fine-tune and add capability to foreign models.²⁶

The Alan Turing Institute said the UK should ask where it can “add value outside the narrow race for the largest foundation model, for example by building capacity around deployment, assurance and systems integration”.⁹¹

However, UK company Cosine announced its intention to release an advanced AI in 2027, which the government said was “Britain’s first fully sovereign frontier AI model” (table 3).

Table 3: Examples of sovereign AI development

Country	Organisation	What have they done?
Singapore	Sapient	Claimed that useful, low-cost, general-purpose models can be made without large amounts of data. ^{92,93}
South Korea	Multiple organisations	Demonstrated their ability to pretrain domestic foundation AI models. ^{i 94,100–104}
UK	Alan Turing Institute	Demonstrated that small models can feasibly be built with near frontier-capability for a specific purpose. ¹⁰⁵
UK	Cosine	Aims to build a fully sovereign AI by the start of 2027 called Lumen Sovereign. It has formed a coalition of UK industry leaders such as BT, HSBC and PwC, to help design the AI model. ¹⁰⁶

In July 2026, the Ada Lovelace Institute recommended that the UK should adopt a ‘fast follower’ strategy, which would use models “that are no longer at the frontier, but which are cheaper and better understood than cutting-edge models”.¹⁰⁷

What assets does the UK have for AI, and what are the challenges?

UK assets for developing AI:

- Datasets for training AI including NHS, BBC and government data.^{26,27,108,109}
- Leading expertise in industries where AI can be applied, such as finance and health and life sciences.^{27,110}
- Research excellence attracting world class talent.^{27,110,111}
- Strong early-stage AI development and startup success.^{112,113}
- Trust due to regulatory stability, democracy and cybersecurity.^{27,110} For example, UK company Wayve, who develop AI for driverless cars, said UK innovation-friendly regulation has allowed them to grow.¹¹¹
- International networks to collaborate on AI,²⁷ for example, the UK hosting of the AI Safety Summit in 2023 and 2025.^{114–116}

ⁱ South Korea also has capability to produce hardware for AI training and has domestic digital platforms to adopt its AI models (for example, search engine Naver is more popular than Google).^{94–99}

Examples of challenges for the UK in AI:

- Protecting AI from being 'poisoned', where published false information can be used to mislead it.^{j 123,124}
- Overcoming high industrial energy prices as AI uses lots of energy.^{k 27,85,127,128}
- Legacy IT systems that may not be able to run the latest software.¹²⁹

^j Some say that Russia publishes false information online for "LLM grooming".^{117–121} Others have said AI may reference disinformation due to "gaps in credible information, rather than foreign interference".¹²²

^k In 2023 UK industrial energy prices were 4 times that of the US and 1.5 times that of France.^{125,126}

2.2

Semiconductor chips

Semiconductor chips are critical to the UK's priority growth sectors, including digital technologies.¹³⁰

The UK Government's 2026 Semiconductor Sector Study said that companies specialising in semiconductors contribute £7.5 billion GVA to the UK economy, and employ 16,350 people.¹³⁰ A 2022 POST briefing said the UK made 0.5% of global semiconductor sales, and that 90% of the chips made in the UK are exported.¹³¹

In July 2026, the Minister for AI said that UK chip production, along with drones, represent the "biggest industrial opportunity of our lifetimes".

Global interdependence in semiconductor chip manufacturing

Silicon is designated a critical mineral in the UK due to supply chain risks and economic importance.¹³² Silicon is the second most abundant element on earth, but refining it is energy and capital intensive.^{133,134} Multiple sources say China has dominated the silicon market, with the Chinese Government providing grants, tax relief and cheap energy to manufacturers, making it more difficult for other countries to compete.^{135–137}

The Taiwan Semiconductor Manufacturing Company (TSMC) is estimated to produce over 90% of the most advanced chips.^{138,139} Production is also dependent on photolithography machines produced by ASML in the Netherlands.^{19,140} ASML's machines require over 900 parts from around the globe.¹⁴¹ Media outlets report that since 2018 the US Government has pressured the Netherlands to not sell this machine to China.^{142–145}

What are the challenges and opportunities in semiconductor chips for the UK?

The UK has strengths in R&D, chip design and intellectual property. It also has a significant global market share of compound semiconductor chips, made of two or more chemical elements such as gallium and nitrogen to make gallium nitride.¹⁴⁶ These have specialised applications such as controlling power (useful for electric vehicles) and transmitting or detecting light (useful for communications, cameras and medical scanners).^{146,147}

The UK has 25 manufacturing facilities for specialised chips, but none for advanced silicon chips.

¹ In the UK, Lochaline Mine in Scotland produces high grade silica sand with very low iron impurity that can be used in chip production.

Challenges for the UK are:^{130,148}

- talent availability; there is a global skills shortage for the semiconductor industry
- energy costs for manufacturers
- access to capital

Some manufacturers said that government support is concentrated on R&D. However, some early-stage R&D firms also report difficulty accessing capital, and raise concerns that success in the UK “tends to result in acquisition rather than sustained independent scaling”.¹³⁰

In 2026, a UK Government report said that the UK has no “sovereign capability” in memory chips.¹³⁰ In 2016, chip architecture designer ARM was sold to SoftBank in Japan.¹⁴⁹ Although the firm remains headquartered in the UK, some MPs said that it should not have been sold.^{150,151}

Some commentators say there is now an opportunity for the UK to develop frontier, next generation chips, including low-power memory chips^m or photonicⁿ chips.^{130,160}

Chip security

Some commentators are concerned about ‘hardware trojans’, [malicious](#) modifications of semiconductor chips.^{161,162} They are challenging to check for. In [telecoms](#) it is recognised that [vulnerabilities in devices can arise in the supply chain](#). In 2026, a media report said that US Intel chips could be vulnerable to cyber-attack and undermine the sovereignty of European datacentres.^o ¹⁶³ Researchers dispute the significance of this vulnerability in practice.¹⁶³

^m Memory chips store data, and the aim of low power memory is to make chips that store data with better energy efficiency. This is important as AI is increasing demand for memory and electricity.^{152,153}

ⁿ[Photonics](#) use light rather than electrons to transmit information and may lead to more efficient and faster computing.^{154–156} In 2026, Cornerstone, the UKs Photonics innovation centre, called for a £305 million investment in a pilot line, a small-scale photonic chip manufacturing line. Its report said that timely investment would allow the UK to commercialise its photonics, build sovereign capability and be a world leader in the area.^{157–159}

^o The media report said that the vulnerability was from ‘management engines’ used to start up intel chips, as they sit outside of the operating system that the user controls.

2.3

Sovereign cloud services

US cloud providers in the UK

Since 2020, UK spending on [the cloud](#) has grown by nearly 30% each year, reaching £10.5 billion in 2024.

US firms Amazon Web Services and Microsoft share 60% to 80% of the UK cloud.^{p 10} In 2025, the Competition and Markets Authority (CMA) concluded that “competition is not working well” in cloud services.^{q 10} Since then, Microsoft and Amazon have made it cheaper and easier to move between cloud providers.^{165,166}

In May 2026, the CMA launched a strategic market status investigation into Microsoft to see if its business software reduces cloud competition. This will conclude in February 2027.¹⁶⁷

Sovereignty as a service

Large-scale cloud providers including US companies Microsoft, Amazon, Oracle and Google, as well as Indian company TATA, offer sovereign cloud services.^{168–172} This is sometimes termed ‘sovereignty as a service’.^{r 22,25,26,173} For example, Microsoft Sovereign Cloud adds encryption keys that are solely managed by the customer, and restricts where data can be stored.

France and Germany have a ‘National Partner Cloud’, where local companies own and operate cloud services that provide Microsoft’s technology.^{174,175}

David Smith, the Chief Digital Adviser to the UK Parliament, evaluated if these sovereign cloud services could protect from either:

1. unwanted data access
2. service withdrawal

He said that customer management of encryption keys means that even if the provider is forced to hand over UK Parliament data (see box 2), it will remain encrypted and unusable for now. No government has claimed to be able to decrypt data protected with the highest level of [encryption](#) (AES-256) without the key.^{s 176} However there is the risk that [quantum computing could](#)

^p This figure is for [infrastructure as a service \(IaaS\)](#), where the cloud provider maintains computational hardware and offers it to customers as remote computing infrastructure.

^q This can be due to effects such as ‘network effects’, where the more users a platform has the better the user experience becomes, and ‘economies of scope’, where additional products can be added using existing expertise and infrastructure at low cost.^{10,18,26,164}

^r ‘Sovereignty as a service’ can also be a term used by critics of large-scale providers of sovereign cloud offerings. They say such services frame sovereignty in a narrow contractual and legal way without providing meaningful control.

^s [Post-quantum cryptographic methods are now recommended](#) by NIST in the US and the National Cyber Security Centre in the UK.

be able to break this encryption eventually, perhaps within the next decade.¹⁷⁷

Service withdrawal is hard to prevent, but some say National Partner Clouds may help. They have not yet been tested in a crisis.¹⁷⁶

Box 2: What is the US CLOUD Act?

The Clarifying Lawful Overseas Use of Data (CLOUD) Act allows the US Government to access data held by US companies outside the US for investigations into serious crime.¹⁷⁸ It came into force in 2018. In 2026, MPs raised concerns about the act in relation to UK reliance on US cloud systems, especially for the NHS and government data.^{5,179–183} The act contains provisions similar to those in other jurisdictions, including the UK’s Crime (Overseas Production Orders) Act 2019.^{184–186}

Table 4: Limits to the US CLOUD Act

Requirements for a successful demand for data ^{185,187}	The relevant data is stored by a US service provider (or US parent company). A warrant, subpoena or court order (as appropriate) has been obtained.
Ways of minimising the impact of a demand	A service provider may challenge and seek to quash or modify a demand in certain limited circumstances¹⁸⁵ Providers cannot be required to have the ability to decrypt data.^{185,186,188}

Despite the limits (table 4), the Parliamentary Office of Speaker’s Counsel said that service providers have to comply with US law (provided a demand has been properly made and all appropriate applications have been followed).¹⁸⁶

How does the UK make use of the US CLOUD Act?

The agreement between the UK and US allows authorities in one country to request data relevant to investigations of serious crime from the other (table 5).¹⁷⁸ The UK has used the CLOUD Act agreement more times than the US (see table 5).

Table 5: How often has the US–UK CLOUD Act agreement been used (2022–24)?

	UK	US
Uses of the agreement	20,142	63
Uses qualifying for mutual legal assistance treaties (MLAT) ^t 191	37	62
Type of use	Mostly real time interception of data under the Investigatory Powers Act 2016	Mostly for stored data, replacing the MLAT process

Source: Data extracted from the 2024 Department of Justice Report to Congress on the renewal of the CLOUD Act agreement between the UK and US.¹⁹²

Government use of Crown Hosting

Crown Hosting is a UK Government cloud service that helps the public sector move away from on-premise facilities and into the public cloud.^{11,193–195}

Crown Hosting can provide data sovereignty (table 6) by storing data in the UK with control of the physical access to datacentres.^u 56,196,197

Table 6: Pillars of information security	
Data confidentiality	Only authorised people can see the data.
Data integrity	Unauthorised edits of data are prevented.
Data availability	The right person can access the data when they need it.

Source: Post interview, Cyber Security News and US Government^{56,198,199}

One POST interviewee said if the UK Government had an issue with a cloud provider, then departments could use Crown Hosting. They said a carefully

^t Mutual legal assistance treaties (MLAT) are a “slow” diplomatic process for requesting stored data, involving foreign governments and courts rather than going directly to the company. The US hoped to replace the burden this placed on them with the CLOUD Act Agreement.^{189,190}

^u Private company Ark owns and operates some UK datacentres and provides data hosting in these datacentres to the public sector via Crown Hosting

managed transition to Crown Hosting would be needed, however, particularly where significant capacity is required.⁵⁶

Cloud procurement options

In 2026, the UK Government said it aimed to coordinate how the public sector buys cloud with the National Cloud Infrastructure Procurement (NCIP) initiative. They said it will provide a stronger position to influence supplier behaviour.^{200,201}

In 2025, the Ministry of Defence bought sovereign cloud capability through Google Cloud in a £400 million deal.^{202–204}

Some UK companies aim to build sovereign cloud, including BT and a Civo partnership with Era4.^{205–207}

2.4

Sovereign timing services

Cloud services need timestamps to update data in the right order.²⁰⁸ Timing is needed for digital systems including finance, telecoms, the power grid and justice.^{209–217} Timing services are considered part of the UK's critical national infrastructure.

Because clocks within digital devices will drift out of sync with each other, external time signals from highly accurate atomic clocks are needed to update internal clocks.^{218–220}

Dependency on Global Navigation Satellite Systems

Timing in the UK largely depends on Global Navigation Satellite Systems (GNSS), a network of satellites with atomic clocks that transmit time signals.²²¹ GNSS includes Galileo from the EU, GLONASS from the Russian Federation, BeiDou from China and GPS from the US.^{222,223} GNSS disruption could cost the UK £1.4 billion per day.^{v 221,224,225}

- Signal jamming and spoofing could be carried out by states, criminals and terrorists.^{w 226–230}
- GNSS operators could switch off signals or fail to maintain satellites due to financial pressure.²²⁶

^v Based on a 2023 estimate, with most of the losses being due to efficiency reduction in road transport and emergency services. Therefore, there could also be life-threatening impacts from not getting ambulances to patients.

^w Jamming is creating a signal at the same frequency as the low-power GNSS signals, preventing the receiver from accurately reading them. Spoofing is deliberately creating a fake signal to trick the receiver into thinking it is at a different position or time.

- Severe space weather^x could disrupt GNSS.²³¹

The National Timing Centre

In 2020, the UK Government set up the National Timing Centre (NTC) at the National Physical Laboratory to provide resilient, sovereign, land-based precision timing signals in the UK as an alternative to GNSS.^{225,232–234}

The NTC said the UK is now a world leader in sovereign timing capability that others look to learn from, although there are still challenges like companies being unwilling to invest to use a sovereign time signal.^{234–236}

2.5

UK Payment System

Payments are increasingly digital.²³⁷ In 2024, only 9% of UK transactions were in cash, and 64% were by debit and credit card.²³⁸ In 2021, US providers Visa and Mastercard handled 95% of UK card payments.²³⁹ The Payment Systems Regulator concluded in 2025 that competition in payment systems is ineffective.²³⁹

In 2014, Russia made a sovereign payment network, NSPK, to avoid US dependency.^y Domestic payments continued when Visa and Mastercard were suspended during the invasion of Ukraine.^{240–246}

The UK Government announced an industry group DeliveryCo in 2025 that will oversee the creation of a domestic payment system by 2030.^{7,247–250}

The UK Government and the Bank of England are also considering the 'digital pound', a digital form of cash that could allow for more account-to-account payments.^{z 253–255}

The UK financial system relies on cloud services. From 13 July 2026, four cloud providers (Microsoft, Amazon, Google and Oracle) have been designated as 'Critical Third Parties' under the Financial Services and Markets Act 2023. Its aim is to improve resilience by giving UK financial regulators oversight of the services provided by these companies to the financial sector to check that suitable resilience planning is in place.^{256,257}

^x Severe space weather includes solar flares and the emission of energetic particles. It now features in the [National Risk Register](#) with impacts including a loss or disruption of GNSS.

^y In 2015 Russia mandated the use of MIR cards that use NSPK for public sector employees. Additionally in 2015 Mastercard and other western companies signed an agreement for their transactions to be processed by NSPK.

^z Account-to-account payments move money directly between bank accounts without the need for intermediaries like credit or debit cards.²⁵¹ Account-to-account payments can be cheaper for merchants.²⁵²

3 How can UK digital sovereignty be increased?

3.1 Open source

Open source software is freely available code that anyone can see, modify, use and distribute.^{aa} It is developed collaboratively by a community, with no individual owning the code.^{bb 261–264}

How does open source support sovereignty?

Open source software could increase sovereignty by:

- allowing local control of software, while benefitting from global support²⁵⁹
- reducing 'vendor lock-in'^{259,265,266}
- giving users increased control over code for changes or auditing^{266,267}

In 2026, the EU said that greater use of open source could support its sovereignty.²⁶⁸

The UK position on open source

The UK is the top contributor of open source code in Europe, and fifth globally.²⁶⁹

The 2025 the UK Government's Technology Code of Practice set criteria to guide UK Government technology, including to "use open source" and being "cloud first".²⁷⁰ Some MPs say that policy on open source is non-binding and inconsistently applied.⁶ Open source software provider Red Hat said that a cloud-first policy is not the same as using open source.²⁵⁹

The non-profit digital campaign organisation Open Rights Group said that open source should be at the centre of a digital sovereignty strategy. They said the government should prefer open source software in procurement, and encourage investment in open source technology with tax incentives.²⁵

^{aa} AI can also be open source if the code, data and key training stage settings and weights are available.²⁵⁸

^{bb} In 2022 the Linux Foundation (non-profit organisation supporting open source development) said open source components typically make up 70% to 90% of most pieces of software.^{259,260}

Security of open-source software and AI

Stakeholders and the UK Government say that open source software is as secure as proprietary software, and sometimes more so.^{259,271,272} However, there is a risk that software is not maintained to fix cybersecurity issues, especially if it is being developed voluntarily. Therefore, even though the software is 'freely available', investment may be necessary to maintain it.²⁵ Some companies sell services to support, test and maintain open source software, such as Red Hat.²⁵⁹

Others are now concerned about the security of open-source AI,²⁷³ as guardrails that cause AI to refuse harmful requests can be removed.²⁷⁴ Some researchers say training models with false data on harmful topics may help, although the makers of ChatGPT say broader investment in resilience to AI threats is needed.^{275,276}

3.2

Procurement

How can procurement be used for UK sovereignty?

Parliamentary committees and the National Audit Office have said that public procurement could do more to support UK digital sovereignty.²² In 2025, the University of Cambridge report Navigating AI Sovereignty said that for UK technology to grow without pressure to move abroad, there needed to be greater domestic demand for products and services (such as from the government).²⁷

As public funds buy £14 billion per year of digital programmes and technology,^{22,27} some stakeholders call on the UK Government to procure digital services from within the UK to build sovereignty.^{25,27,31,32,58}

In 2025, a Ministry of Defence contract^{cc} with US company Palantir was awarded without tender, causing MPs and media reports to ask whether the UK had become locked into dependency on the company.^{13,277–279} Palantir was also awarded a £330 million contract in 2023 to provide a data platform for the NHS.^{11,280} In 2026, the Science, Innovation and Technology Committee said that the UK Government should use the 2027 break clause in the Palantir contract and develop a UK-based data platform.^{dd 11,12}

Procurement can also set standards for software to be interoperable, meaning that systems can connect and communicate in a standard way with little effort from the user, reducing vendor lock-in.^{18,26,281–284}

^{cc} The contract was to provide data analytics for strategic, tactical and live operational decision making.

^{dd} This is to "safeguard against the risk of vendor lock-in" and because Palantir shows a "clear mismatch with UK values" since they supply software "supporting highly controversial policies and activities" in the US and its co-founder "has criticised the concept of a national health service".

Some stakeholders say allies should align procurement standards for greater influence on technology providers.^{18,282}

What would enable sovereign procurement?

In a POST interview, techUK said that the Procurement Act 2023^{ee} could allow procurement decisions based on sovereignty, but in practice they are usually made based on lowest cost.^{58,285} Andy Burnham said in 2026 that the UK has “chased cut-price deals around the world” and should instead use procurement to support British-based suppliers.^{286–288} Some MPs have called for a sovereign procurement strategy.²⁸⁹

In response to a parliamentary question in 2026, the UK Government said:^{20,290}

Procurement decisions are taken on the basis of value for money, security and the effective delivery of public services.

techUK said companies need to know how they would be assessed against sovereignty.⁵⁸ In 2026, Collin Bannon of BT said the equivalent of a “Red Tractor mark”^{ff} should be introduced to show how sovereign products are.⁵⁹

3.3

Building skills for digital sovereignty

In 2025, Public Digital said that it is important to have “skills, processes and knowledge” for digital sovereignty.³⁰ Commentators say that digital and data skills gaps could undermine sovereignty.^{291,292}

How many jobs will require digital skills?

In 2026, the UK Government projected 239,000 new priority jobs and 249,000 replacement roles in the UK digital and technologies sector between 2025 and 2035.²⁹³

The UK Government’s 2025 AI Opportunities Action Plan said that tens of thousands of AI professionals need to be trained by 2030. It also said that accurate assessment of the AI skills gap is a priority as “current estimates are imprecise and outdated”.²⁹⁴

^{ee} Government guidance says “The overall basis of award is now referred to as the ‘most advantageous tender’ (MAT), rather than, as in the previous legislation, the ‘most economically advantageous tender’ (MEAT). However, this is not a change in policy and the change is to clarify and reinforce for contracting authorities that tenders do not have to be awarded on the basis of lowest price/cost, or that price/cost must always take precedence over non price/cost factors”.²⁸⁵

^{ff} The “red tractor” mark, equivalent to that used for food and drinks, would allow customers to check claims of sovereignty against a standard. It could help justify premium costs for more sovereign products.

Could a lack of skills undermine sovereignty?

In 2026, former Chief Scientific Adviser for National Security, Anthony Finkelstein, said that the UK will need sovereign skills before building sovereign capabilities.²⁹⁵ He said that the UK needs to maintain “ongoing analysis” of skills vulnerabilities to avoid the “slow erosion of capacity”,²⁹⁶ and ensure that the UK has the [skills to buy, check, operate and fix critical technologies](#).²⁹²

For example, the UK has traditionally been strong at radio-frequency engineering, which is critical for defence, and position, navigation and timing (PNT) such as GNSS, but now has an ageing workforce, few UK national students, and a lack of funding.²⁹⁶

In 2025, the European Commission said [digital skills must be improved](#) to engage citizens in digital sovereignty efforts.⁴⁰

In 2026, the UK Government announced [apprenticeship units](#), short courses available in England for “critical skills areas such as digital, advanced AI and engineering”. It also launched free AI courses with the [AI Skills Boost Programme](#).

The need for digital skills in governance

In a POST interview, the executive director of the Open Rights Group, Jim Killock, suggested having an internal consultancy in government to support digital sovereignty.^{99 25,272}

In 2025, Public Digital said that the UK needs leaders “who grasp how digital systems work” to make informed decisions about sovereignty.³⁰ The UK Government’s 2025 guidance for Audit and Risk Assurance Committees, which support a company board by reviewing risk management, financial statements, and annual reports, said digital skills in the committee would be beneficial if digital systems are important to the organisation.³⁰⁰

3.4

Mapping out digital systems

In 2026, Anthony Finkelstein said the UK needs an evidence base from which to build a digital sovereignty strategy that prioritises investment and intervention. This evidence could take the form of “resilience maps” for technology that identify:²⁹²

- critical systems

⁹⁹ During the POST interview Mr Killock mentioned DINUM and ZenDis were examples of internal government consultants.²⁷² DINUM supports the French state to develop its digital strategy, aiming to make the state more efficient and sovereign through digital technology.²⁹⁷ It is similar to the Government Digital Service (GDS) in the UK, although GDS does not have an explicit focus on sovereignty.²⁹⁸ ZenDis is the “central state actor for digital sovereignty in Germany”.²⁹⁹

- global supply dependencies
- single points of failure
- credible disruption scenarios

European Commission briefings also say that the EU needs to map and understand existing dependencies.^{291,301}

The need to map digital systems may also apply for companies. The UK 2025 Government Cyber Governance Code of Practice, which supports boards and directors to govern cyber risks, suggested that companies identify and prioritise “critical technology, information and services”.³⁰²

The Tony Blair Institute has said there is a need for “structured engagement” between the government and the private sector, as companies will have earlier and better insight into vulnerabilities and emerging areas where the UK can gain advantage. It said this needs to go beyond “ad hoc consultation”.¹⁸

Others say there are organisations with convening power that may help build resilience maps and digital sovereignty strategy, such as the National Physical Laboratory and telecoms providers.^{234,303,304}

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